



Redefining Musical Success: Chart Longevity

1. Introduction

1.1 Background and Motivation

Hit Song Science (HSS) is an interdisciplinary field that emerged with advances in Music Information Retrieval (MIR) and the availability of large-scale music data, examining whether musical success can be explained or predicted using quantitative analysis of audio features, metadata, and social signals (Downie, 2003; Seufitelli et al., 2023). Research in HSS typically relies on large datasets and statistical or machine learning models, driven by both academic interest and commercial demand from record labels and digital platforms (Yap & Raheem, 2024).

Musical success remains an important object of research because music markets are characterised by extreme inequality and uncertainty; a small fraction of songs captures a disproportionate share of attention and revenue, while most receive little sustained engagement (Ferreira & Waldfogel, 2012; Salganik et al., 2006). These skewed outcomes make success difficult to anticipate yet economically consequential, sustaining debate over its predictability, particularly given the role of social reinforcement (Pachet & Roy, 2008; Salganik et al., 2006). As a result, musical popularity provides an informative setting for studying how success emerges in cultural markets (Ferreira & Waldfogel, 2012).

1.2. Literature Review

Recent reviews document convergence in how musical success is defined and modelled within HSS (Seufitelli et al., 2023; Yap & Raheem, 2024). Most work operationalises success using static chart outcomes, such as chart entry or predefined chart rank thresholds, and frames prediction as a binary task (Interiano et al., 2018; Pachet & Roy, 2008; Zhao et al., 2023). Other research measures success using peak chart position (Lee and Lee, 2018), while a smaller subset adopts continuous indicators such as streaming or play counts (Yang et al., 2017; Saragih, 2023). Across these approaches, engineered acoustic features dominate model inputs; however, reported associations with success are mixed, ranging from moderate effects to weak or negligible relationships, with findings being sensitive to outcome definition and dataset choice (Lee & Lee, 2018; Saragih, 2023). By contrast, artist history, prior popularity, and exposure-related variables exhibit more robust associations with success, and several studies report improved predictive performance when such factors are incorporated (Seufitelli et al., 2023; Interiano et al., 2018; Zhao et al., 2023).

A central limitation of the HSS literature is its emphasis on prediction over explanation, as prediction-dominant approaches risk diminishing returns in the absence of explanatory insight. Success is commonly operationalised as a binary outcome, despite broad recognition that musical success is temporal and cumulative (Salganik et al., 2006; Seufitelli et al., 2023; Yap & Raheem, 2024). A single chart appearance captures initial exposure but collapses variation between transient attention and sustained success, treating moderately successful songs and long-lasting hits as equivalent (Lee & Lee, 2018; Pachet & Roy, 2008; Seufitelli et al., 2023). As a result, different models may achieve similar predictive performance across datasets or feature sets, yet without explanatory approaches, it remains unclear which relationships are structural,

which reflect measurement artefacts, and which generalise beyond specific contexts (Seufitelli et al., 2023; Yap & Raheem, 2024).

This limitation is particularly problematic given that prior work conceptualises musical success as a dynamic, socially driven process. Cultural success is stochastic and socially amplified, with early popularity reinforced through feedback loops that increase visibility and adoption (Salganik et al., 2006). Song popularity follows diffusion-like trajectories of growth, peak, and decline that vary systematically (Rosati et al., 2021). From this perspective, chart presence functions as a cumulative signal of popularity that static success measures fail to capture.

Despite this process-based understanding, chart longevity is rarely modelled directly, leaving persistence largely unexplored (Seufitelli et al., 2023; Yap & Raheem, 2024). This omission is particularly limiting because chart-based success is capacity-constrained and inherently comparative, reflecting competition for limited audience attention (Mayerl et al., 2023). Duration-based measures, such as chart longevity, therefore provide a more informative indicator of sustained success than one-time chart outcomes. Lee and Lee (2018) show that longevity constitutes a distinct and substantively meaningful dimension of success, with most songs exhibiting short chart runs and only a small minority persisting for many weeks. As a result, binarising success collapses critical variation in duration, treating transient and sustained chart presence as equivalent.

1.3 Research Aim

This research aims to reframe musical success as a duration-based process by modelling chart longevity to provide explanatory insight into cumulative advantage and social reinforcement mechanisms operating after chart entry (see Table 1).

Table 1

Research Questions Guiding Analysis

Research Question	Description
RQ1	To what extent do artist history, prior popularity, and exposure-related factors explain sustained chart presence?
RQ2	What role do song-level characteristics play in shaping sustained chart presence, conditional on artist and exposure factors?

2. Methods

2.1 Data Collection

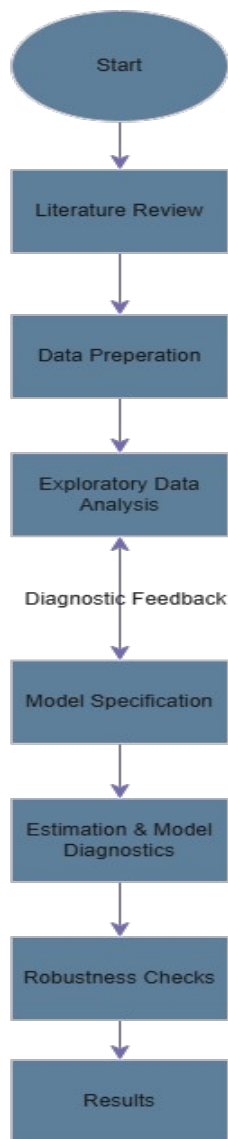
This study uses MusicOSet, an open dataset of artists, songs, and albums (Silva et al., 2019). MusicOSet is well-suited to this analysis because it provides longitudinal chart histories alongside artist-level prior exposure and song-level characteristics, allowing musical success to be modelled as a cumulative duration process rather than a binary or peak-based outcome. The dataset is publicly available at <https://marianaossilva.github.io/DSW2019/index.html>.

2.2 Research Pipeline

This study adopts a diagnostic-driven, explanatory data science approach in which data preparation and exploratory analysis iteratively inform variable construction and model specification, consistent with standard data science workflows (Komorowski et al., 2016; Schutt & O’Neil, 2013). Figure 1 summarises the analytical workflow, highlighting the role of exploratory analysis in model development.

Figure 1

Analytical Workflow and Research Pipeline



All data processing, visualisation, and statistical analyses were conducted in R (version 4.5.1). Packages used in the analysis are listed in Table 2.

Table 2*R Packages Used for Data Processing, Visualisation, and Statistical Analysis*

Package	Version	Role in analysis
tidyverse	2.0.0	Data ingestion, cleaning, reshaping, exploratory analysis, and visualisation
scico	1.5.0	Colour-blind safe diverging colour scales for correlation heatmap visualisation
MASS	7.3-65	Negative Binomial regression modelling
AER	1.2-15	Overdispersion testing for Poisson models
performance	0.15.3	Collinearity diagnostics for regression models
broom	1.0.11	Tidying and formatting model outputs
DHARMa	0.4.7	Simulation-based residual diagnostics

Note. Analyses were conducted in R using RStudio. Package versions correspond to the computational environment used for the final analyses.

2.3 Data Preparation

This study adopts an explanatory approach to HSS by modelling chart longevity as a distinct dimension of musical success. Acoustic features were retained for comparability with prior HSS research despite mixed findings (Seufitelli et al., 2023; Yap & Raheem, 2024), while cumulative advantage and social reinforcement were operationalised through artist-history and exposure-related IVs, consistent with theories of socially reinforced success (Interiano et al., 2018; Salganik et al., 2006).

Only song-level mergeable dataframes containing a week variable were retained to ensure temporal alignment at chart entry and avoid post-run contamination; consequently, unmergeable variables (e.g. artist follower counts) were excluded. This yielded three datasets: song-level acoustic features ($N = 20,405$), song metadata containing artist credits and song type ($N = 20,405$), and weekly chart observations ($N = 250,392$). Weekly observations were aggregated to the song level to ensure independence and avoid pseudo-replication (Hurlbert, 1984), yielding 20,402 songs.

Songs entering the dataset mid-run were excluded to avoid left-censoring of chart longevity and entry-rank measures, excluding 98 songs (Allison, 2014). Weeks-on-chart values equal to zero were recoded to one to reflect chart-entry conventions. Chart longevity was defined as the maximum number of weeks on chart, and entry position as the rank in the first observed chart week.

Artist-history variables were constructed using only songs charting prior to each song's entry, enforcing strict temporal ordering. These captured cumulative experience (number of prior songs and collaborating artists), typical chart persistence (average weeks on chart across prior songs), and typical peak performance (average peak rank across prior songs). For songs with multiple credited artists, artist-history values were averaged to reflect shared cumulative advantage. Acoustic features and metadata, invariant across weekly observations, were retained as single song-level values. One song contained invalid acoustic feature values and was dropped, yielding a final analytical sample of 20,303 songs.

2.4 Analysis

2.4.1 Exploratory Data Analysis (EDA)

EDA followed the data discovery framework of Komorowski et al. (2016), using distributional inspection, predictor diagnostics, and functional-form checks to inform model specification prior to estimation, consistent with best practice for count-data analysis (Cameron & Trivedi, 2013).

The DV was examined using histograms and summary statistics to assess discreteness, skewness, overdispersion, and near-zero inflation. IVs were summarised to evaluate scale, spread, plausibility, and category balance. Pairwise correlations among continuous predictors were assessed using a correlation heatmap to identify potential multicollinearity. Bivariate DV–IV relationships were examined using binned mean plots to assess monotonicity and

compatibility with a log-link function, while categorical-variable distributions were inspected to assess sparsity and guide category collapsing and reference-level selection.

2.4.2 Modelling

Guided by EDA diagnostics, chart longevity was modelled using a log-linked count regression to ensure non-negative fitted values and multiplicative effects (Cameron & Trivedi, 2013). A Poisson baseline was estimated to assess equidispersion, motivating Negative Binomial (NB) regression as the primary estimator for overdispersed counts (Hilbe, 2011). The main specification jointly modelled artist-history and song-level acoustic variables, with an extended specification including musical categoricals estimated as a sensitivity analysis to assess robustness to sparsely populated and imbalanced controls identified during EDA (Angrist & Pischke, 2009). Continuous predictors were standardised (1 SD) to improve numerical stability and comparability, categorical predictors were factor-encoded, and diagnostics included checks of design-matrix rank and Cook's distance to assess identifiability and influence (Cook, 1977; Pettit & Belsley, 1992; Gelman, 2008).

2.5 Answering Research Questions

The research questions focus on explaining variation in chart longevity rather than predicting hit status; therefore, classification- and prediction-oriented methods were unsuitable, as they do not permit interpretable inference about persistence. Given that chart longevity remains underexplored in HSS and evidence on acoustic features is mixed, a diagnostic-driven explanatory approach was adopted. A regression framework enables estimation of conditional

associations between artist history, song-level acoustic features, and chart persistence. NB regression was used to accommodate overdispersed count outcomes and yield multiplicative effects consistent with cumulative-advantage theory, allowing both RQ1 and RQ2 to be addressed jointly.

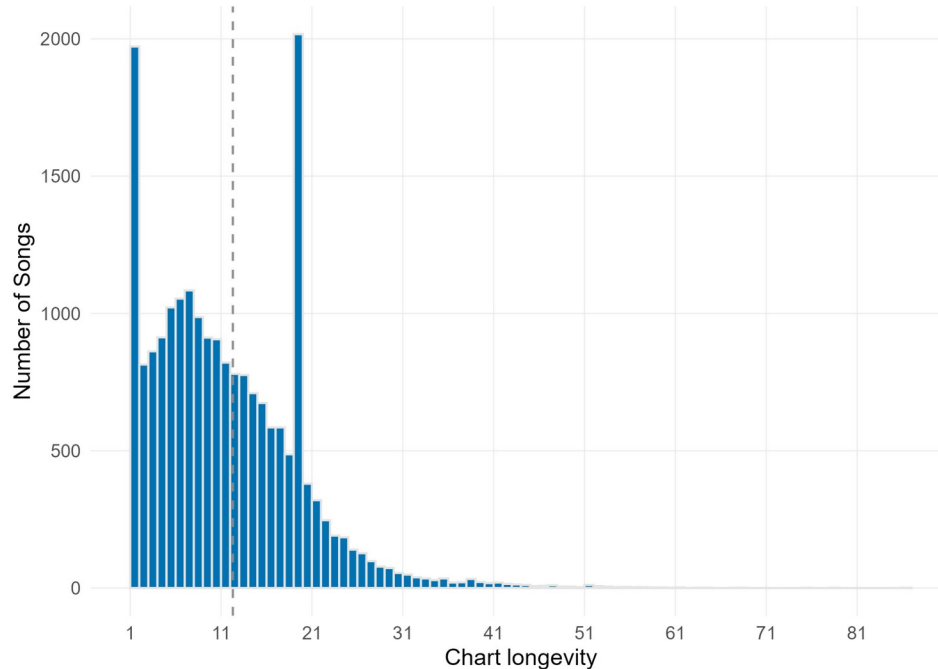
3. Results

3.1 Exploratory Data Analysis

EDA examined the distribution of the outcome variable, predictor structure, and potential functional relationships. Chart longevity exhibited a right-skewed count distribution with substantial overdispersion ($M = 12.3$ weeks; variance = 67.59 weeks²), violating the Poisson equidispersion assumption (Figure 2). Only 5.3% of songs remained on the chart for one week, and 9.7% for two weeks or fewer, providing no evidence of structural zero inflation (Cameron & Trivedi, 2013).

Figure 2

Distribution of Chart Longevity



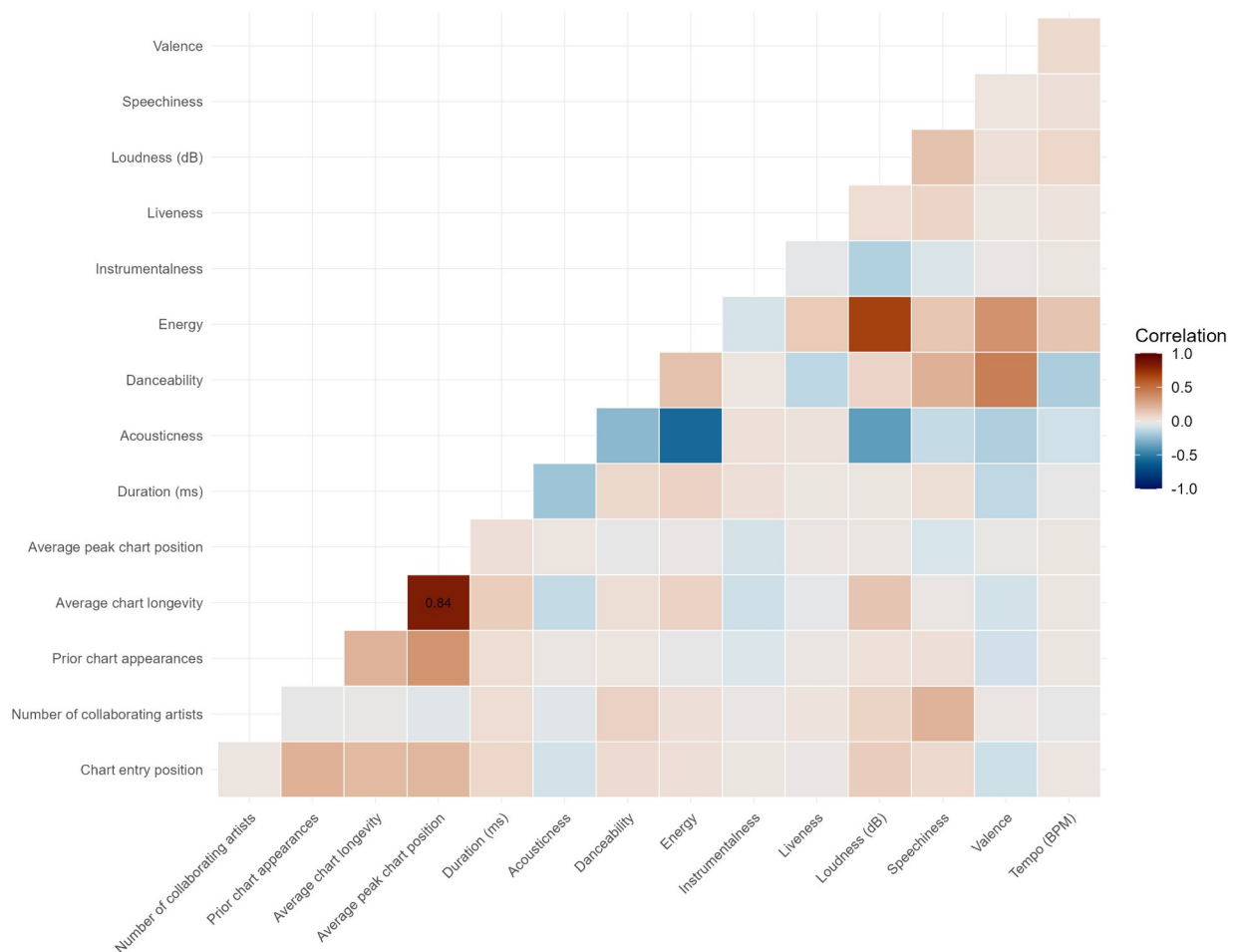
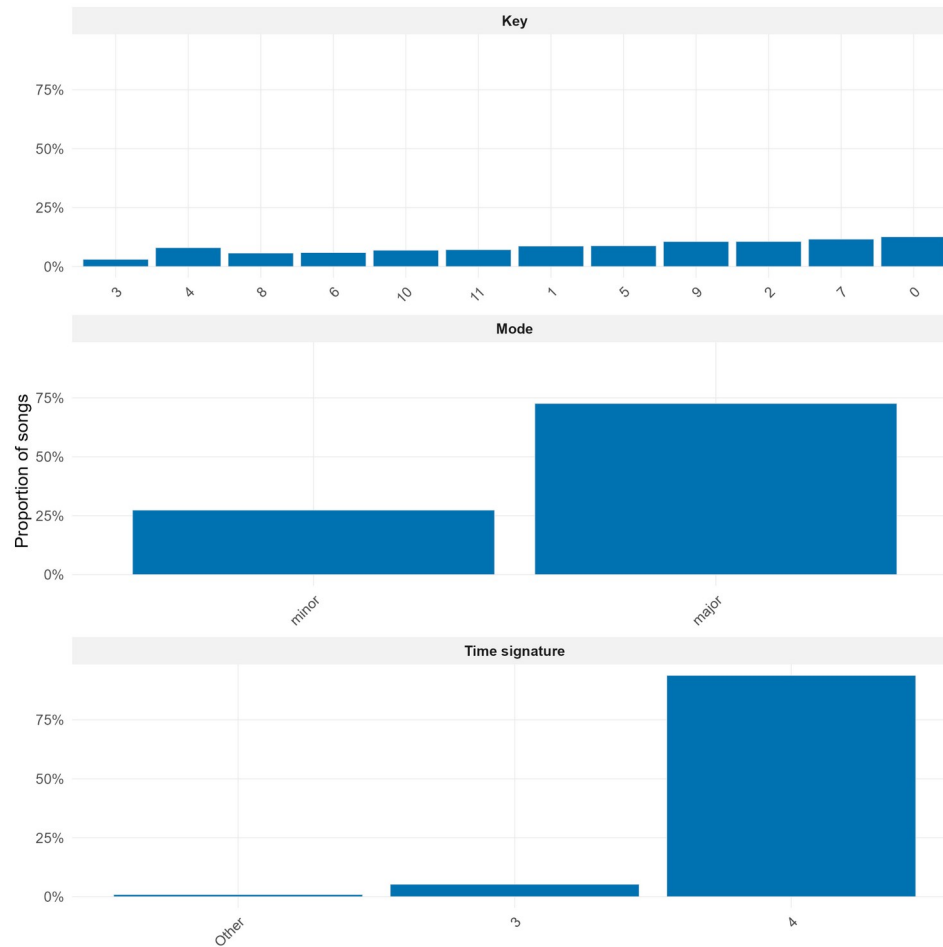


Figure 4

Distribution of Musical Categorical Variables



Binned mean plots were used to assess functional form. Relationships between continuous predictors and chart longevity were generally smooth, with no evidence of threshold effects that would motivate non-linear or piecewise specification; minor irregularities reflected sparse support rather than structural non-linearity (Figure 5) (Harrell, 2015). Chart longevity distributions showed substantial overlap across song types (Figure 6), suggesting limited evidence of strong group separation at the exploratory stage. Together, these findings supported

the use of a log-linear count-regression framework for subsequent modelling (Cameron & Trivedi, 2013).

Figure 5
Binned Mean Outcomes vs Core Predictors

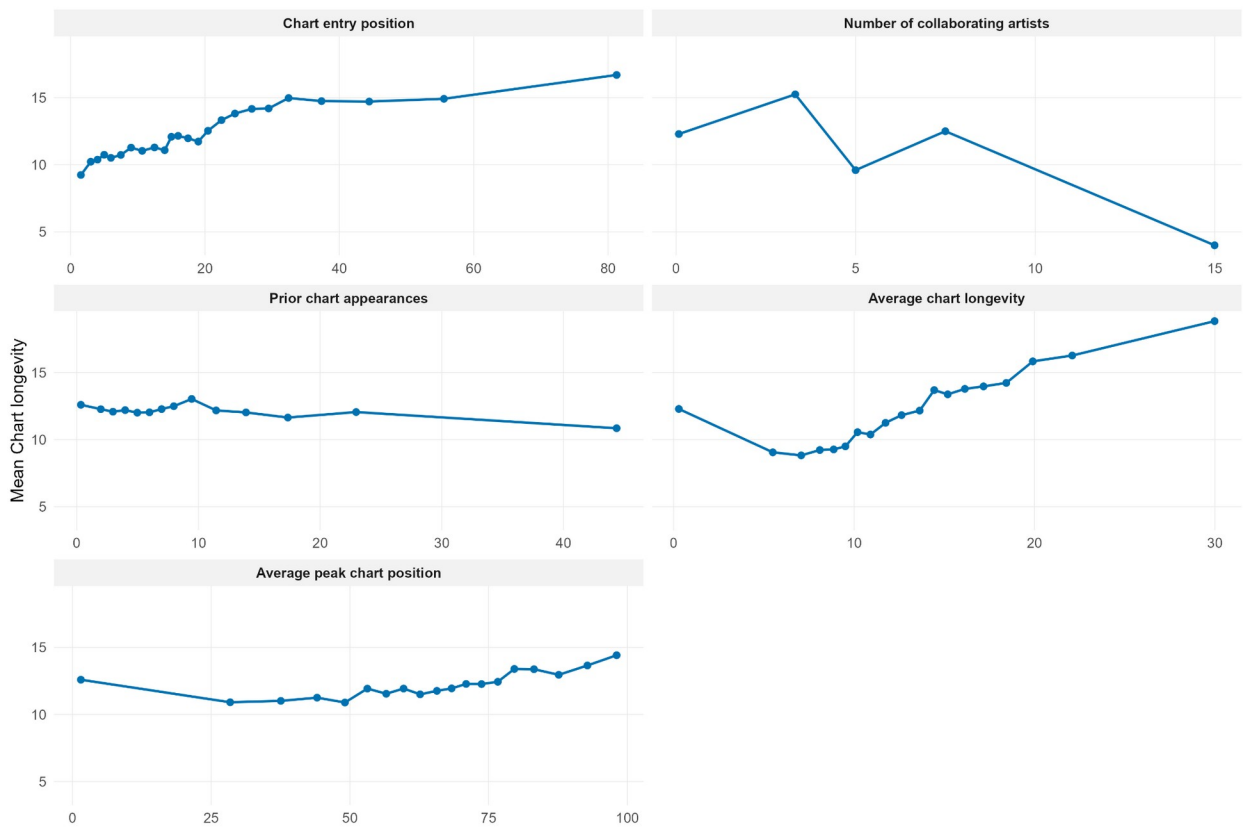
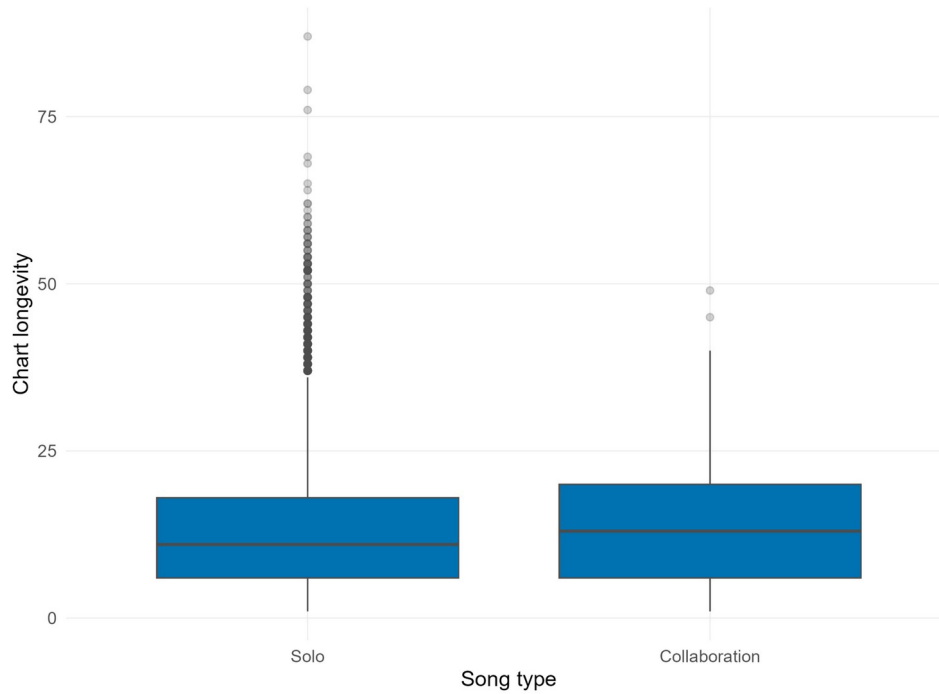


Figure 6
Chart Longevity by Collaboration Type



3.2 Descriptive Statistics

The final analytical sample comprised 20,303 songs. Chart longevity ranged from 1 to 87 weeks ($Mdn = 11$, $IQR = 12$). The sample included 5,992 artists, of whom 3,088 were debut artists. Descriptive statistics for predictor variables are reported in Table 3.

Table 3

Descriptive Statistics for Predictor Variables

Variable	M (SD)	Mdn (IQR)	Range
Pre-entry factors			
Chart entry position	20.57 (18.17)	16.00 (19.00)	1.00–100.00
Prior chart appearances	6.35 (10.05)	3.00 (7.78)	0.00–109.00

Average chart longevity	10.09 (8.00)	10.12 (14.90)	0.00–67.00
Average peak chart position	47.88 (33.06)	56.50 (73.48)	0.00–100.00
Number of collaborating artists	0.09 (0.39)	0.00 (0.00)	0.00–15.00
Acoustic features			
Duration (ms)	229,892 (66,899)	223,373 (72,873)	24,604–1,561,133
Danceability	0.60 (0.15)	0.61 (0.20)	0.10–0.99
Acousticness	0.26 (0.26)	0.17 (0.40)	0.00–0.99
Energy	0.63 (0.20)	0.64 (0.30)	0.01–1.00
Instrumentalness	0.05 (0.17)	0.00 (0.00)	0.00–0.98
Liveness	0.19 (0.16)	0.13 (0.16)	0.01–1.00
Loudness (dB)	−8.84 (3.59)	−8.46 (5.17)	−30.90–3.44
Speechiness	0.07 (0.07)	0.04 (0.03)	0.02–0.95
Tempo (BPM)	120.33 (27.67)	118.96 (36.15)	37.11–231.03
Valence	0.61 (0.24)	0.63 (0.38)	0.03–1.00

Note. *M* = mean; *SD* = standard deviation; *Mdn* = median; *IQR* = interquartile range.

3.3 Model Fitting

Given the non-negative, overdispersed count nature of the outcome, chart longevity was modelled using NB regression with a log link, estimated via maximum likelihood. A Poisson model with an identical predictor set was estimated for diagnostic comparison. Dispersion testing indicated substantial overdispersion in the Poisson specification (dispersion statistic = 51.34, $p < .001$), and the NB model provided a markedly superior fit (AIC: 135,461 vs. 176,914).

A robustness specification incorporating musical categorical variables (key, mode, and time signature) yielded only marginal improvement in fit ($AIC = 135,436$) and did not materially alter estimated effects. Given the robustness of results and the sparsity of categorical levels, the parsimonious NB model excluding these variables was retained for main reporting (Harrell, 2015).

Moderate collinearity was observed between artist prior longevity ($VIF = 4.09$) and average peak rank ($VIF = 4.29$), reflecting conceptual relatedness but remaining below conventional thresholds. Adjusted VIFs were approximately 2 with tolerance values above .20, and estimates were stable across specifications; both variables were retained to capture distinct dimensions of prior artist success (Dormann et al., 2012; Harrell, 2015). Influence diagnostics indicated that estimates were not driven by individual observations (maximum Cook's $D = 0.009$), supporting the adequacy of the final model specification (Cook, 1977; Pettit & Belsley, 1992).

3.4 Negative Binomial Regression

To examine factors associated with chart longevity and explain musical success, NB regression models were estimated. The overall model was statistically significant, $\chi^2(15) = 55.16$, $p < .001$, indicating that the included predictors jointly explain significantly more variation in chart longevity than an intercept-only model. Table 4 reports incidence rate ratios.

Table 4*Negative Binomial Regression of Chart Longevity*

Predictor	β	SE	IRR	95% CI for IRR	<i>p</i>
Pre-entry factors					
Chart entry position	0.129	0.005	1.14	[1.13, 1.15]	< .001***
Prior chart appearances	−0.046	0.005	0.96	[0.95, 0.97]	< .001***
Average chart longevity	0.253	0.009	1.29	[1.27, 1.31]	< .001***
Average peak chart position	−0.223	0.009	0.80	[0.79, 0.82]	< .001***
Number of collaborating artists	0.011	0.012	1.01	[0.99, 1.04]	.338
Acoustic features					
Duration (ms)	0.079	0.005	1.08	[1.07, 1.09]	< .001***
Acousticness	−0.046	0.006	0.95	[0.94, 0.97]	< .001***

Danceability	0.072	0.006	1.07	[1.06, 1.09]	< .001***
Energy	−0.033	0.008	0.97	[0.95, 0.98]	< .001***
Instrumentalness	−0.021	0.005	0.98	[0.97, 0.99]	< .001***
Liveness	−0.022	0.005	0.98	[0.97, 0.99]	< .001***
Loudness (dB)	0.034	0.007	1.03	[1.02, 1.05]	< .001***
Speechiness	−0.040	0.005	0.96	[0.95, 0.97]	< .001***
Valence	−0.021	0.006	0.98	[0.97, 0.99]	< .001***
Tempo (BPM)	0.010	0.005	1.01	[1.00, 1.02]	.030*

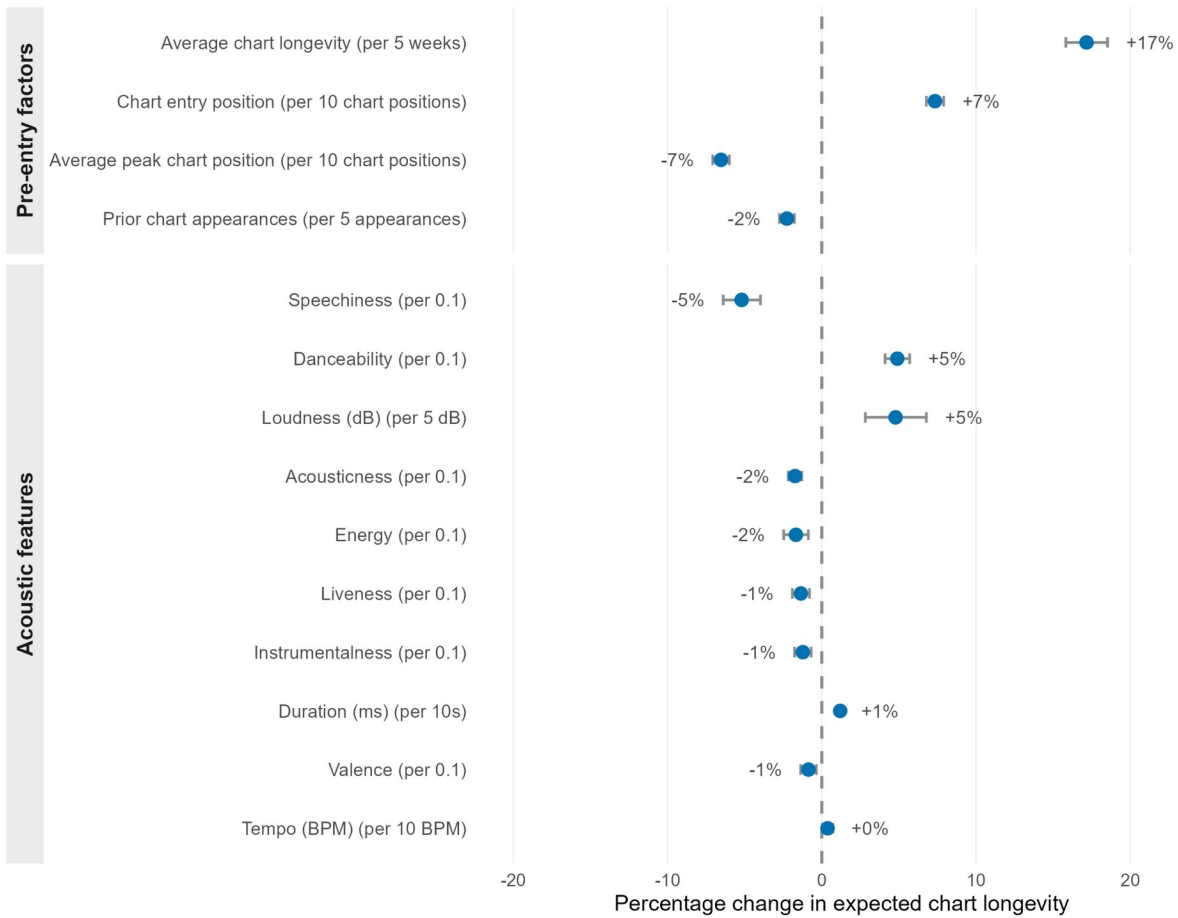
Note. Continuous predictors were standardised (1 SD), except for the number of collaborating artists due to its discrete count interpretation. IRRs, therefore, represent the multiplicative change in expected chart duration per one standard deviation increase.

Significance levels: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Significant results were rescaled post-estimation for interpretability (Figure 7). Effects are reported for empirically meaningful unit changes that are small relative to the observed variability of each predictor (e.g. fractions of an IQR or SD in the final modelling sample), ensuring that effect magnitudes correspond to realistic changes within the data rather than arbitrary scaling choices (Cameron & Trivedi, 2013; Gelman, 2007).

Figure 7

Percent Change in Chart Longevity for Statistically Significant Predictors



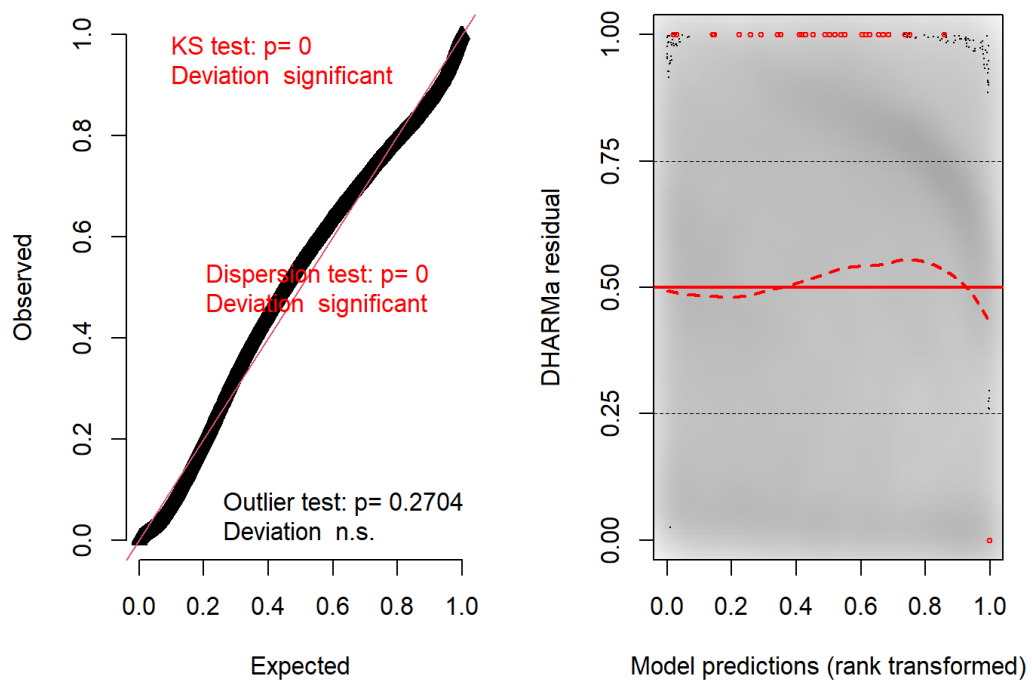
Note. All effects are reported as percentage changes in expected chart longevity for meaningful unit increases and should be interpreted as proportional rather than additive.

Results indicate that chart longevity is primarily associated with artist pre-entry characteristics, while song-level acoustic features contribute statistically detectable but comparatively minor variation once these factors are controlled for. Model diagnostics support the reliability of these estimates: DHARMA simulation-based diagnostics were used to assess model adequacy (Hartig, 2016). The KS and dispersion tests were significant, consistent with large-sample sensitivity, while the outlier test was non-significant ($p = .27$). Residual-fitted and

QQ plots showed no systematic patterns, supporting adequate functional form and variance specification (Figure 8).

Figure 8

DHARMA Residual Diagnostics for the Negative Binomial Model



4. Discussion

4.1 Answering Research Questions

4.1.1 Artist History and Exposure (RQ1)

Artist history and exposure-related factors explain a substantial proportion of variation in sustained chart presence, with artist history emerging as the dominant component. Prior average chart longevity exhibits the strongest association with persistence, indicating that sustained prior success is far more predictive of future chart duration than other historical or exposure-based measures. Entry visibility also contributes positively, suggesting that early exposure facilitates subsequent persistence, but its influence is clearly secondary to accumulated artist success, consistent with debut strength facilitating rather than determining sustained success. In contrast, peak-oriented measures of prior popularity and repeated chart exposure are negatively associated with longevity, indicating that transient prominence and repeated appearances do not translate into sustained chart presence and may reflect front-loaded or saturating attention rather than durable popularity.

4.1.2 Acoustic Features (RQ2)

Song-level characteristics play a limited but non-negligible role in shaping sustained chart presence once artist history and exposure factors are controlled for. Acoustic features are statistically associated with chart longevity, but their effects are clearly secondary to artist pre-entry characteristics. Features linked to contemporary, studio-oriented production, such as higher danceability and loudness, are associated with slightly greater persistence, while speech-dominant, acoustic, live, or instrumental tracks tend to exhibit shorter chart runs. Other attributes, including tempo and duration, contribute little substantively, indicating that song-level

characteristics condition chart longevity at the margin rather than acting as primary drivers of sustained success.

4.2 Relation to Existing Research

Framing musical success as a longevity-based outcome reveals artist history as a central driver of sustained chart presence. This aligns with social influence models of musical success in which outcomes are socially driven and dynamically reinforced after initial exposure, with visibility-based feedback loops generating cumulative advantage in persistence rather than selection alone (Salganik et al., 2006). At the same time, the comparatively smaller role of chart entry exposure indicates that early visibility facilitates, but does not determine, sustained success, consistent with process-based views of popularity as dynamic rather than front-loaded (Rosati et al., 2021).

However, prior artist history does not uniformly translate into sustained success. Repeated exposure can exhibit diminishing returns, as visibility does not scale linearly into durable consumption and may instead reflect saturation effects in attention-constrained music markets (Ferreira & Waldfogel, 2012; Salganik et al., 2006). This pattern suggests that cumulative advantage underpins the durability of consumer demand by operating selectively through historically sustained success rather than uniformly across exposure-related mechanisms, with collaborative breadth offering little reinforcement of long-term chart persistence (Mayerl et al., 2023).

Patterns observed in song-level acoustic features are more plausibly interpreted as reflections of mainstream exposure-related effects and genre attribution rather than intrinsic determinants of success. Prior research suggests that formatting conventions help explain weak associations with track duration (Lee & Lee, 2018; Saragih, 2023), while broader acoustic patterns align with evidence that contemporary charting songs increasingly conform to tightly produced, studio-oriented norms characterised by high loudness and rhythmic engagement. Variability in HSS findings that include acoustic features may therefore reflect shifts in dominant genres over time, as genres cluster around distinct acoustic profiles (Pachet & Roy, 2008; Saragih, 2023). These patterns align with sceptical perspectives in the HSS and MIR literature that acoustic descriptors often proxy genre affiliation and production practices rather than musical quality itself, pointing to stylistic convergence in mainstream production rather than strong causal mechanisms of chart success (Downie, 2003; Interiano et al., 2018; Pachet & Roy, 2008).

Taken together, these findings highlight limitations in dominant binary and peak-based measures of success in HSS (Seufitelli et al., 2023; Yap & Raheem, 2024). By conflating short-lived chart prominence with sustained persistence, such measures obscure empirically distinct dynamics of transient attention and durable popularity (Lee & Lee, 2018; Pachet & Roy, 2008).

5. R Code, GitHub Pages

Full project code is available on GitHub (see Table 5). Project structure is shown in Figures 9 and 10, with the project README displayed in Figure 11.

Table 5

GitHub Profile and Project Repository

Resources	URL
GitHub Profile	https://github.com/ArchieIrving
Project Repository	https://github.com/ArchieIrving/hit-song-science

Figure 9

Structure of the GitHub Repository

```
hit-song-science/
├── R/
│   ├── 00_run_all.R
│   ├── 01_data_cleaning.R
│   ├── 02_eda.R
│   ├── 03_fit_nb_models.R
│   ├── 04_report_nb_outputs.R
│   ├── analysis_setup.R
│   └── helpers.R
├── raw/
│   ├── features/
│   │   └── acoustic_features.csv
│   ├── metadata/
│   │   └── songs.csv
│   ├── popularity/
│   │   └── song_chart.csv
├── clean/
│   └── song_df.csv
└── outputs/
    └── descriptives/
```

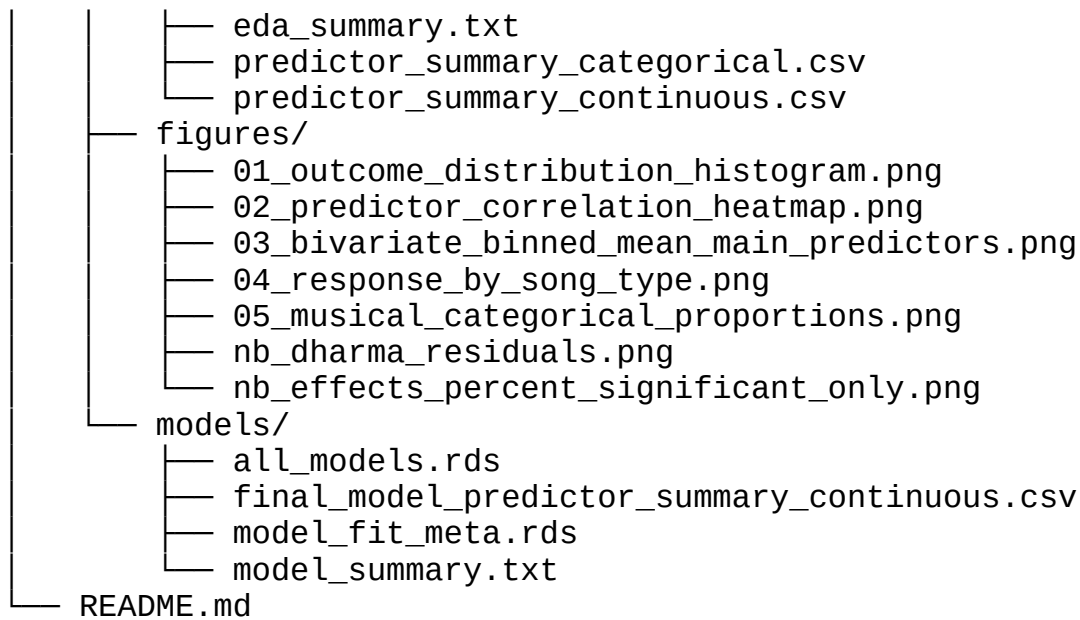


Figure 10

GitHub Project Landing Page

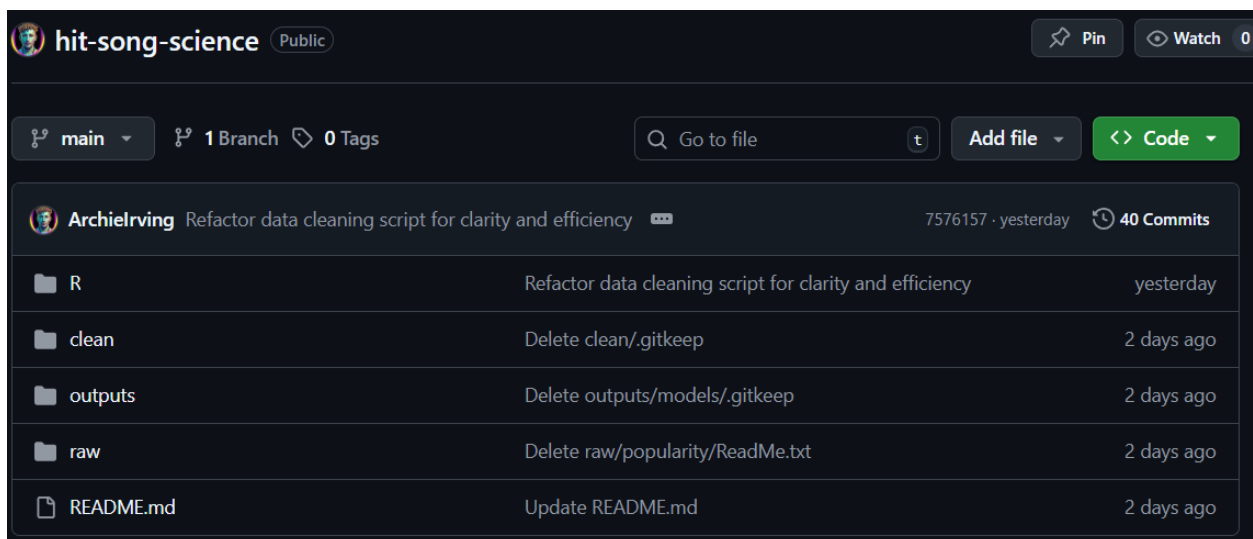


Figure 11

Project [README.md](#) Available on the GitHub Project Page

Modelling Song Chart Longevity

Project Overview

The aim of this project was to extend research in *Hit Song Science* by adopting an explanatory modelling approach to musical success. Rather than treating success as a binary outcome (chart entry versus non-entry), the study operationalised success as chart longevity, measured by the number of weeks a song remained on the *Billboard* charts.

Data and Methods

The analysis used data from MusicOSet, an open, curated dataset integrating *Billboard* chart performance with song-level metadata and Spotify-derived acoustic features. The dataset contains 20,405 songs from the U.S. popular music industry spanning 1962–2018.

To examine sustained chart presence, a Negative Binomial regression model was estimated to address two research questions:

- **RQ1:** To what extent do artist history, prior popularity, and exposure-related factors explain sustained chart presence?
- **RQ2:** What role do song-level characteristics play in shaping sustained chart presence, conditional on artist and exposure factors?

Key Findings

The results indicate that artist prior chart longevity is the strongest positive predictor of weeks on chart, while entry rank exhibits a smaller but positive association. In contrast, prior peak chart position and repeated chart exposure are negatively associated with chart longevity, suggesting diminishing returns to peak-oriented or repeated success.

Song-level acoustic features were statistically significant but exhibited **modest effect sizes**. Higher danceability and loudness were positively associated with chart longevity, whereas greater speechiness, acousticness, liveness, instrumentalness, and energy were negatively associated. Overall, the findings reinforce long-standing conclusions in *Hit Song Science* that audio features play a limited explanatory role relative to artist history and exposure dynamics.

How to Run the Analysis

Requirements

- R (version 4.x)
- R packages: tidyverse, MASS, AER, performance, DHARMa, broom

How to Run the Analysis

1. Clone or download this repository to your local machine and open the project in R or RStudio.
2. Set the project root directory as your working directory. The project root is the top-level folder containing the `R/`, `raw/`, `clean/`, and `outputs/` directories.
3. Ensure all required R packages are installed (see the package loading statements at the top of the scripts).
4. Run the full analysis pipeline from the project root using:

```
source("R/00_run_all.R")
```



Running `00_run_all.R` executes the entire analysis pipeline end-to-end. This includes data cleaning and aggregation.

6. Conclusion

This project adopted an explanatory approach to underexplored dimensions of HSS by operationalising musical success as chart longevity. This enabled an examination of how social and exposure-related factors, particularly artist history and chart entry position, shape sustained chart performance alongside song-level acoustic characteristics.

6.1 Reflections

6.1.1 Key Findings

- Artist history emerged as the strongest predictor of chart longevity, exceeding the influence of song-level acoustic features and highlighting cumulative advantage and prior exposure.
- Chart entry position showed a secondary but meaningful association with longevity, facilitating persistence without determining long-term success.
- Acoustic features showed statistically significant but modest effects, operating primarily at the margins relative to social and exposure-related factors.
- Joint acoustic patterns suggested genre-linked persistence, with more danceable, louder, but less speech-dominant acoustic, live, and instrumental tracks exhibiting longer chart durations, consistent with mainstream, studio-oriented genres.
- Chart longevity captured a distinct dimension of musical success beyond binary or peak-based measures, offering an alternative perspective on HSS.

6.1.2 Limitations and Assumptions

Limitations

- Chart longevity captures relative chart endurance rather than absolute consumption and may be sensitive to platform and culture-specific contexts, limiting generalisability.
- Genre was not explicitly modelled, rendering links between acoustic features and genre speculative.
- The observational design restricts causal inference, as estimated effects reflect associations rather than experimentally isolated relationships.
- Unobserved industry factors, including label promotion, radio programming, were not directly measured.

Assumptions

- Chart longevity is a valid indicator of sustained musical success.
- Artist history variables accurately capture prior exposure at the point of chart entry.
- An NB model is appropriate for modelling overdispersed chart-duration data.

6.1.3 Future Work

- Compare charting and non-charting songs and examine determinants of chart entry position to better distinguish drivers of initial exposure from sustained success.
- Disaggregate acoustic effects by genre to test whether observed longevity patterns reflect genre-specific dynamics.
- Replicate the longevity framework across periods to assess stability under changing industry conditions.

6.2 Engagement

Engagement with employability and industry events reshaped my understanding of data science practice. Industry speakers emphasised the importance of communication and collaboration, particularly translating complex analyses for non-technical stakeholders and working across multidisciplinary teams, challenging my assumption that technical ability alone determines effectiveness and prompting me to actively develop and foreground these skills.

A guest lecture by Clive Humby further highlighted how large-scale data-driven decision-making can amplify both positive and harmful social outcomes, underscoring data scientists' responsibility to balance technical impact with ethical judgment and accountability. This prompted a more critical awareness of how analytical choices shape real-world consequences and has influenced how I approach data analysis, particularly in being more deliberate about assumptions, framing, and downstream implications.

References

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Appendix

Appendix A

Full Project R Code

analysis_setup.R

```
# R/analysis_setup.R -----
# Analysis definitions, variable sets, and labelling conventions.

suppressPackageStartupMessages({
  library(dplyr)
  library(withr)
  library(stringr)
  library(tibble)
})

withr::local_options(list(tibble.width = Inf, scipen = 999))

# ---- Core constants -----

OUTCOME_VAR <- "weeks_on_chart"

# ---- Variable sets (match clean/song_df.csv) -----

core_vars <- c(
  "song_rank_entry",
  "n_artists_collab",
  "artist_prior_n_songs",
  "artist_avg_longevity_prior",
  "artist_avg_peak_rank_prior"
)

audio_vars <- c(
  "duration_ms",
  "acousticness", "danceability", "energy", "instrumentalness",
  "liveness", "loudness", "speechiness", "valence", "tempo"
)

cat_vars <- c(
  "key", "mode", "time_signature", "song_type"
)
```

```
# ---- Labels -----
```

```
NICE_NAMES <- c(
  song_rank_entry      = "Chart entry position",
  n_artists_collab     = "Number of collaborating artists",
  artist_prior_n_songs = "Prior chart appearances",
  artist_avg_longevity_prior = "Average chart longevity",
  artist_avg_peak_rank_prior = "Average peak chart position",
  duration_ms         = "Duration (ms)",
  loudness             = "Loudness (dB)",
  tempo               = "Tempo (BPM)",
  acousticness        = "Acousticness",
  danceability        = "Danceability",
  energy              = "Energy",
  instrumentality     = "Instrumentality",
  liveness            = "Liveness",
  speechiness         = "Speechiness",
  valence             = "Valence",
  key                 = "Key",
  mode                = "Mode",
  time_signature      = "Time signature",
  song_type           = "Song type",
  weeks_on_chart      = "Chart longevity"
)
```

```
label_from_map <- function(x, mapping = NICE_NAMES) {
  x_chr <- as.character(x)
  dplyr::recode(x_chr, !!!as.list(mapping), .default = x_chr)
}
```

```
named_labels <- function(vars, mapping = NICE_NAMES) {
  vars <- as.character(vars)
  setNames(label_from_map(vars, mapping), vars)
}
```

```
# ---- Encoding -----
```

```
encode_factors <- function(df) {
  stopifnot(is.data.frame(df))
  stopifnot(all(cat_vars %in% names(df)))
```

```
df |>
  dplyr::mutate(
    # Spotify "key" uses -1 for unknown
    key = dplyr::na_if(key, -1),
    key = factor(key),
```

```

mode = factor(mode, levels = c(0, 1), labels = c("minor", "major")),
mode = stats::relevel(mode, ref = "minor"),

song_type = factor(song_type, levels = c("Solo", "Collaboration")),
song_type = stats::relevel(song_type, ref = "Solo"),

time_signature = as.character(time_signature),
time_signature = dplyr::if_else(time_signature %in% c("3", "4"), time_signature, "Other"),
time_signature = factor(time_signature, levels = c("4", "3", "Other"))
)
}

# ---- Scaling -----

standardise_predictors <- function(df, core_vars, audio_vars, exclude = character()) {
  scale_vars <- intersect(c(core_vars, audio_vars), names(df))
  scale_vars <- setdiff(scale_vars, exclude)

  df_scaled <- df |>
  dplyr::mutate(
    dplyr::across(dplyr::all_of(scale_vars), ~ as.vector(scale(.x)))
  )

  list(
    df_scaled = df_scaled,
    scale_vars = scale_vars
  )
}

# ---- Effect-size increments -----

INC_MAP <- tibble::tibble(
  term = c(
    "song_rank_entry",
    "artist_prior_n_songs",
    "artist_avg_longevity_prior",
    "artist_avg_peak_rank_prior",
    "duration_ms",
    "tempo",
    "loudness",
    "acousticness",
    "danceability",
    "energy",
    "instrumentalness",
    "liveness",

```



```

    "speechiness",
    "valence"
  ),
  inc = c(
    10, 5, 5, 10,
    10000, 10, 5,
    0.10, 0.10, 0.10, 0.10, 0.10, 0.10, 0.10
  )
)

label_increment_suffix <- function(term) {
  dplyr::case_when(
    term == "song_rank_entry"      ~ "(per 10 chart positions)",
    term == "artist_prior_n_songs" ~ "(per 5 appearances)",
    term == "artist_avg_longevity_prior" ~ "(per 5 weeks)",
    term == "artist_avg_peak_rank_prior" ~ "(per 10 chart positions)",
    term %in% c(
      "acousticness", "danceability", "energy", "instrumentalness",
      "liveness", "speechiness", "valence"
    ) ~ "(per 0.1)",
    term == "tempo" ~ "(per 10 BPM)",
    term == "loudness" ~ "(per 5 dB)",
    term == "duration_ms" ~ "(per 10s)",
    TRUE ~ ""
  )
}

# ---- Formula builder -----

build_nb_formula <- function() {
  rhs <- c(core_vars, audio_vars, cat_vars)
  stats::as.formula(paste0(OUTCOME_VAR, " ~ ", paste(rhs, collapse = " + ")))
}

```

helpers.R

```

# R/helpers.R -----
# Small helper utilities shared across the analysis pipeline:
# - directory creation
# - saving plots / text / objects
# - a project-wide ggplot theme

```

```

suppressPackageStartupMessages({
  library(ggplot2)
  library(readr)
})

# ---- Directories -----

ensure_dirs <- function(paths) {
  stopifnot(is.character(paths), length(paths) >= 1)

  paths <- unique(paths)

  for (p in paths) {
    if (dir.exists(p)) next
    dir.create(p, showWarnings = FALSE, recursive = TRUE)
  }

  invisible(TRUE)
}

# ---- Saving helpers -----

# Write text output to disk, creating directories if needed.
write_lines_safe <- function(lines, path) {
  stopifnot(is.character(path), length(path) == 1)
  stopifnot(is.character(lines))

  ensure_dirs(dirname(path))
  readr::write_lines(lines, path)

  invisible(TRUE)
}

# Save an R object as .rds, ensuring the output directory exists.
save_rds_safe <- function(object, path) {
  stopifnot(is.character(path), length(path) == 1)

  ensure_dirs(dirname(path))
  saveRDS(object, path)

  invisible(TRUE)
}

# Save a ggplot with consistent export settings.
save_plot <- function(filename,
  plot_obj,

```

```

        dir = "outputs/figures",
        width = 7,
        height = 5,
        dpi = 300,
        units = "in") {
stopifnot(is.character(filename), length(filename) == 1)
stopifnot(is.character(dir), length(dir) == 1)

ensure_dirs(dir)

ggplot2::ggsave(
  filename = file.path(dir, filename),
  plot = plot_obj,
  width = width,
  height = height,
  units = units,
  dpi = dpi
)

invisible(TRUE)
}

# ---- Plot theme -----

# Project palette / styling constants
COL_PRIMARY <- "#0072B2"
COL_NEUTRAL <- "grey30"
COL_GRID <- "grey92"
COL_OUTLINE <- "grey90"
COL_REF_LINE <- "grey55"

theme_project <- function(base_size = 12) {
  ggplot2::theme_minimal(base_size = base_size) +
  ggplot2::theme(
    plot.title = ggplot2::element_text(face = "bold", colour = COL_NEUTRAL),
    plot.title.position = "plot",
    panel.grid.minor = ggplot2::element_blank(),
    panel.grid.major = ggplot2::element_line(colour = COL_GRID, linewidth = 0.25),
    axis.text = ggplot2::element_text(colour = COL_NEUTRAL),
    strip.text = ggplot2::element_text(face = "bold"),
    strip.background = ggplot2::element_rect(fill = "grey95", colour = NA)
  )
}

```

00_run_all.R

```
# R/00_run_all.R -----
# Runs the full analysis pipeline end-to-end from the project root.

#All scripts assume the working directory is set to the parent folder of the project root (i.e. the
directory containing raw/, clean/, outputs/, and R/).

options(tibble.width = Inf, scipen = 999)

run_ts <- Sys.time()
cat("=====\n")
cat("Pipeline run:", format(run_ts), "\n")
cat("=====\n\n")

source("R/helpers.R")
ensure_dirs(c("clean", "outputs/figures", "outputs/descriptives", "outputs/models"))

cat("[1/4] Running data cleaning (R/01_data_cleaning.R)...\n")
source("R/01_data_cleaning.R")
cat("[1/4] Done.\n\n")

cat("[2/4] Running EDA (R/02_eda.R)...\n")
source("R/02_eda.R")
cat("[2/4] Done.\n\n")

cat("[3/4] Fitting models (R/03_fit_nb_models.R)...\n")
source("R/03_fit_nb_models.R")
cat("[3/4] Done.\n\n")

cat("[4/4] Generating reporting outputs (R/04_report_nb_outputs.R)...\n")
source("R/04_report_nb_outputs.R")
cat("[4/4] Done.\n\n")

elapsed <- difftime(Sys.time(), run_ts, units = "secs")

cat("=====\n")
cat("Pipeline completed successfully.\n")
cat("Elapsed time (seconds): ", round(as.numeric(elapsed), 1), "\n", sep = "")
cat("=====\n")
```

01_data_cleaning.R

```
# R/01_data_cleaning.R -----
# Construct a song-level dataset for modelling chart longevity.
# Output: clean/song_df.csv

suppressPackageStartupMessages({
  library(dplyr)
  library(tidyr)
  library(readr)
  library(stringr)
  library(purrr)
})

source("R/helpers.R")
source("R/analysis_setup.R")
ensure_dirs("clean")

# ---- Load inputs -----

acoustic_features <- read_tsv("raw/features/acoustic_features.csv", show_col_types = FALSE)
song_chart <- read_tsv("raw/popularity/song_chart.csv", show_col_types = FALSE)
songs <- read_tsv("raw/metadata/songs.csv", show_col_types = FALSE)

stopifnot(anyDuplicated(acoustic_features$song_id) == 0)

# ---- Restrict to songs observed from chart entry -----

# Keep songs observed from chart entry (avoids left-truncated chart runs).
valid_song_ids <- song_chart |>
  transmute(
    song_id,
    week,
    weeks_on_chart = as.integer(weeks_on_chart)
  ) |>
  group_by(song_id) |>
  summarise(
    entry_week = min(week, na.rm = TRUE),
    entry_weeks_on_chart = weeks_on_chart[which.min(week)],
    .groups = "drop"
  ) |>
  filter(!is.na(entry_weeks_on_chart), entry_weeks_on_chart %in% c(0L, 1L)) |>
  pull(song_id)

song_chart <- filter(song_chart, song_id %in% valid_song_ids)
songs <- filter(songs, song_id %in% valid_song_ids)
```

```

acoustic_features <- filter(acoustic_features, song_id %in% valid_song_ids)

# ---- Parse credited artists from metadata -----

# Extract Spotify artist IDs from the artists dictionary field.
parse_artists_dict <- function(song_id_vec, artists_str_vec) {
  map2_dfr(song_id_vec, artists_str_vec, function(sid, s) {
    if (is.na(s) || !nzchar(s)) {
      return(tibble(song_id = sid, artist_id = NA_character_))
    }

    m <- str_match_all(s, "([A-Za-z0-9]{22})\\s*:")[[1]]

    if (nrow(m) == 0) {
      return(tibble(song_id = sid, artist_id = NA_character_))
    }

    tibble(song_id = sid, artist_id = m[, 2])
  })
}

songs_artists_long <- parse_artists_dict(songs$song_id, songs$artists) |>
  filter(!is.na(artist_id)) |>
  distinct(song_id, artist_id)

artist_lists <- songs_artists_long |>
  group_by(song_id) |>
  summarise(
    n_artists = n_distinct(artist_id),
    artist_id = paste(unique(artist_id), collapse = ";"),
    .groups = "drop"
  ) |>
  mutate(n_artists_collab = pmax(n_artists - 1L, 0L))

# ---- Song-level chart summaries -----

song_chart_summary <- song_chart |>
  transmute(
    song_id,
    week,
    # entry week is sometimes recorded as 0; recode to 1 for consistency
    weeks_on_chart = if_else(weeks_on_chart == 0, 1L, as.integer(weeks_on_chart)),
    rank_score
  ) |>
  group_by(song_id) |>
  summarise(

```

```

    song_entry_week = min(week, na.rm = TRUE),
    song_longevity = max(weeks_on_chart, na.rm = TRUE),
    song_rank_peak = max(rank_score, na.rm = TRUE),
    .groups = "drop"
  )

song_rank_entry <- song_chart |>
  transmute(song_id, week, rank_score) |>
  left_join(
    song_chart_summary |> dplyr::select(song_id, song_entry_week),
    by = "song_id"
  ) |>
  filter(week == song_entry_week) |>
  group_by(song_id) |>
  summarise(song_rank_entry = max(rank_score, na.rm = TRUE), .groups = "drop")

song_chart_summary <- song_chart_summary |>
  left_join(song_rank_entry, by = "song_id") |>
  transmute(
    song_id,
    song_entry_week,
    weeks_on_chart = song_longevity,
    song_rank_peak,
    song_rank_entry
  )

# ---- Artist prior chart history (strictly prior songs) -----

# Prior-only artist history: lagged cumulative averages prevent outcome leakage.
artist_song <- songs_artists_long |>
  left_join(song_chart_summary, by = "song_id") |>
  arrange(artist_id, song_entry_week, song_id)

artist_song_prior <- artist_song |>
  group_by(artist_id) |>
  mutate(
    prior_n_songs = row_number() - 1L,

    cum_longevity = cumsum(weeks_on_chart),
    artist_avg_longevity_prior_artist =
      if_else(prior_n_songs > 0,
        lag(cum_longevity) / prior_n_songs,
        0),

    cum_peak_rank = cumsum(song_rank_peak),
    artist_avg_peak_rank_prior_artist =

```

```

    if_else(prior_n_songs > 0,
            lag(cum_peak_rank) / prior_n_songs,
            0)
  ) |>
ungroup() |>
dplyr::select(
  song_id,
  artist_id,
  prior_n_songs,
  artist_avg_longevity_prior_artist,
  artist_avg_peak_rank_prior_artist
)

song_artist_history <- artist_song_prior |>
group_by(song_id) |>
summarise(
  artist_prior_n_songs = mean(prior_n_songs),
  artist_avg_longevity_prior = mean(artist_avg_longevity_prior_artist),
  artist_avg_peak_rank_prior = mean(artist_avg_peak_rank_prior_artist),
  .groups = "drop"
) |>
mutate(
  artist_avg_longevity_prior = round(artist_avg_longevity_prior, 2),
  artist_avg_peak_rank_prior = round(artist_avg_peak_rank_prior, 2)
)

# ---- Assemble final song-level dataset -----

song_df <- songs |>
dplyr::select(song_id, song_name, song_type) |>
left_join(
  song_chart_summary |> dplyr::select(song_id, weeks_on_chart, song_rank_entry),
  by = "song_id"
) |>
left_join(
  artist_lists |> dplyr::select(song_id, artist_id, n_artists_collab),
  by = "song_id"
) |>
left_join(
  song_artist_history |>
  dplyr::select(
    song_id,
    artist_prior_n_songs,
    artist_avg_longevity_prior,
    artist_avg_peak_rank_prior
  ),

```



```

    by = "song_id"
  ) |>
  left_join(
    acoustic_features |>
      dplyr::select(
        song_id, duration_ms, key, mode, time_signature,
        acousticness, danceability, energy, instrumentalness,
        liveness, loudness, speechiness, valence, tempo
      ),
    by = "song_id"
  ) |>
  mutate(tempo = na_if(tempo, 0)) |>
  dplyr::select(
    song_id, song_name, song_type,
    weeks_on_chart,
    song_rank_entry,
    artist_id, n_artists_collab,
    artist_prior_n_songs,
    artist_avg_longevity_prior, artist_avg_peak_rank_prior,
    duration_ms, key, mode, time_signature,
    acousticness, danceability, energy, instrumentalness,
    liveness, loudness, speechiness, valence, tempo
  )

# Drop observations with missing modelling variables
model_cols <- c(OUTCOME_VAR, core_vars, audio_vars, cat_vars)

song_df <- song_df |>
  drop_na(all_of(model_cols))

# ---- Final checks and save -----

stopifnot(anyDuplicated(song_df$song_id) == 0)
stopifnot(all(song_df$weeks_on_chart >= 1))
stopifnot(all(song_df$weeks_on_chart == round(song_df$weeks_on_chart)))

write_csv(song_df, "clean/song_df.csv")
message("Data cleaning complete: clean/song_df.csv")

```

02_eda.R

```
# R/02_eda.R -----
```

```

# Exploratory analysis for chart longevity (weeks_on_chart).
# Inputs: clean/song_df.csv
# Outputs: outputs/figures/*.png; outputs/descriptives/*.csv; eda_summary.txt

suppressPackageStartupMessages({
  library(tidyverse)
  library(scales)
  library(rlang)
  library(scico)
})

source("R/analysis_setup.R")
source("R/helpers.R")

ensure_dirs(c("outputs/figures", "outputs/descriptives"))

# ---- Correlation heatmap (continuous predictors) -----

plot_corr_heatmap <- function(df, vars, label_cutoff = 0.70) {
  vars <- intersect(vars, names(df))
  stopifnot(length(vars) >= 2)

  df_num <- df |>
    dplyr::select(all_of(vars)) |>
    mutate(across(everything(), as.numeric))

  cmat <- stats::cor(df_num, use = "pairwise.complete.obs")

  cor_long <- as.data.frame(cmat) |>
    tibble::rownames_to_column("var1") |>
    pivot_longer(-var1, names_to = "var2", values_to = "corr") |>
    mutate(
      var1 = factor(var1, levels = vars),
      var2 = factor(var2, levels = vars),
      i = as.integer(var1),
      j = as.integer(var2)
    ) |>
    # avoid duplicating mirrored correlations
    filter(j > i) |>
    mutate(
      # label only materially large correlations
      label = if_else(
        !is.na(corr) & abs(corr) >= label_cutoff,
        sprintf("%.2f", corr),
        ""
      )
    )
}

```

```

)

axis_labels <- named_labels(vars)

ggplot(cor_long, aes(x = var2, y = var1, fill = corr)) +
  geom_tile(colour = "white", linewidth = 0.3) +
  geom_text(aes(label = label), size = 3) +
  scale_fill_scico(
    palette = "vik",
    limits = c(-1, 1),
    breaks = c(-1, -0.5, 0, 0.5, 1),
    name = "Correlation"
  ) +
  scale_x_discrete(labels = axis_labels) +
  scale_y_discrete(labels = axis_labels) +
  labs(x = NULL, y = NULL) +
  theme_project() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1, size = 9),
    axis.text.y = element_text(size = 9),
    panel.grid = element_blank()
  )
}

# ---- Main EDA routine -----

run_eda <- function(df,
  outcome_var = OUTCOME_VAR,
  core_vars = character(),
  audio_vars = character(),
  cat_vars = character(),
  main_cat = c("song_type"),
  n_bins = 25) {

  cont_all <- intersect(c(core_vars, audio_vars), names(df))
  cat_all <- intersect(cat_vars, names(df))
  deep_cont <- intersect(core_vars, names(df))
  deep_cat <- intersect(main_cat, names(df))

  # ---- Outcome distribution -----

  y <- df[[outcome_var]]

  p_outcome <- ggplot(df, aes(x = .data[[outcome_var]])) +
    geom_histogram(
      binwidth = 1,

```

```

    boundary = 0,
    colour = COL_OUTLINE,
    fill = COL_PRIMARY
  ) +
  geom_vline(
    # mean/variance contrast is used later to motivate NB over Poisson
    xintercept = mean(y, na.rm = TRUE),
    linetype = "dashed",
    colour = COL_REF_LINE
  ) +
  scale_x_continuous(
    breaks = seq(1, max(y, na.rm = TRUE), by = 10),
    limits = c(1, NA)
  ) +
  labs(
    x = label_from_map(outcome_var),
    y = "Number of songs"
  ) +
  theme_project()

save_plot("01_outcome_distribution_histogram.png", p_outcome)

# ---- Correlation heatmap -----

if (length(cont_all) >= 2) {
  save_plot(
    "02_predictor_correlation_heatmap.png",
    plot_corr_heatmap(df, cont_all),
    width = 11,
    height = 8.5
  )
}

# ---- Predictor summaries -----

if (length(cont_all) > 0) {
  cont_tab <- df |>
  summarise(
    across(
      all_of(cont_all),
      list(
        mean = ~ mean(., na.rm = TRUE),
        sd = ~ sd(., na.rm = TRUE),
        median = ~ median(., na.rm = TRUE),
        min = ~ min(., na.rm = TRUE),
        max = ~ max(., na.rm = TRUE),

```

```

    p25  = ~ quantile(., 0.25, na.rm = TRUE),
    p75  = ~ quantile(., 0.75, na.rm = TRUE)
  ),
  .names = "{.col}__{.fn}"
)
)|>
pivot_longer(everything(), names_to = "stat", values_to = "value") |>
separate(stat, into = c("variable", "stat"), sep = "__") |>
pivot_wider(names_from = stat, values_from = value) |>
mutate(
  IQR = p75 - p25,
  Variable = label_from_map(variable)
) |>
dplyr::select(Variable, mean, sd, median, min, max, IQR) |>
arrange(Variable)

write_csv(cont_tab, "outputs/descriptives/predictor_summary_continuous.csv")
}

if (length(cat_all) > 0) {
  cat_tab <- df |>
  dplyr::select(all_of(cat_all)) |>
  pivot_longer(everything(), names_to = "variable", values_to = "category") |>
  drop_na(category) |>
  mutate(
    Variable = label_from_map(variable),
    category = as.character(category)
  ) |>
  count(Variable, category, name = "count") |>
  group_by(Variable) |>
  mutate(proportion = count / sum(count)) |>
  ungroup() |>
  arrange(Variable, desc(count))

  write_csv(cat_tab, "outputs/descriptives/predictor_summary_categorical.csv")
}

# ---- Musical categorical proportions -----

mus_cat <- intersect(c("key", "mode", "time_signature"), names(df))

if (length(mus_cat) > 0) {
  df_props <- df |>
  dplyr::select(all_of(mus_cat)) |>
  pivot_longer(everything(), names_to = "variable", values_to = "category") |>
  drop_na(category) |>

```

```

mutate(
  variable = factor(variable, levels = c("time_signature", "mode", "key")),
  variable_label = label_from_map(variable),
  category = as.character(category)
) |>
count(variable, variable_label, category, name = "count") |>
group_by(variable, variable_label) |>
mutate(proportion = count / sum(count)) |>
ungroup() |>
group_by(variable) |>
mutate(category = forcats::fct_reorder(category, proportion)) |>
ungroup()

p_props <- ggplot(df_props, aes(x = category, y = proportion)) +
  geom_col(fill = COL_PRIMARY, colour = COL_OUTLINE, linewidth = 0.2) +
  scale_y_continuous(labels = percent_format(accuracy = 1)) +
  facet_wrap(~ variable_label, scales = "free_x", ncol = 1) +
  labs(x = NULL, y = "Proportion of songs") +
  theme_project() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

save_plot("05_musical_categorical_proportions.png", p_props, width = 9, height = 9)
}

# ---- Binned mean outcome by core predictors -----

if (length(deep_cont) > 0 && outcome_var %in% names(df)) {
  make_quantile_breaks <- function(x, n_bins) {
    brks <- unique(quantile(x, probs = seq(0, 1, length.out = n_bins + 1), na.rm = TRUE))
    # quantile breaks can collapse when predictors have few unique values
    if (length(brks) < 5) brks <- pretty(range(x, na.rm = TRUE), n = 10)
    brks
  }

  df_long <- df |>
  dplyr::select(all_of(c(deep_cont, outcome_var))) |>
  pivot_longer(cols = all_of(deep_cont), names_to = "predictor", values_to = "x") |>
  mutate(y = .data[[outcome_var]]) |>
  filter(is.finite(x), is.finite(y))

  d_binned <- df_long |>
  group_by(predictor) |>
  group_modify(function(d, key) {
    # inspect functional form without imposing linearity
    brks <- make_quantile_breaks(d$x, n_bins)

```

```

d |>
  mutate(bin = cut(x, breaks = brks, include.lowest = TRUE)) |>
  group_by(bin) |>
  summarise(
    x_mid = mean(x, na.rm = TRUE),
    mean_y = mean(y, na.rm = TRUE),
    .groups = "drop"
  ) |>
  filter(is.finite(x_mid), is.finite(mean_y))
}) |>
ungroup() |>
mutate(
  predictor_label = label_from_map(predictor),
  predictor_label = factor(predictor_label, levels = label_from_map(deep_cont))
)

p_binned <- ggplot(d_binned, aes(x = x_mid, y = mean_y)) +
  geom_line(linewidth = 0.8, colour = COL_PRIMARY) +
  geom_point(size = 1.8, colour = COL_PRIMARY) +
  facet_wrap(~ predictor_label, scales = "free_x", ncol = 2) +
  labs(
    x = NULL,
    y = paste0("Mean ", label_from_map(outcome_var))
  ) +
  theme_project()

save_plot("03_bivariate_binned_mean_main_predictors.png", p_binned, width = 12, height
= 8)
}

# ---- Outcome by category -----

if (length(deep_cat) > 0) {
  for (v in deep_cat) {
    p_cat <- ggplot(df, aes(x = .data[[v]], y = .data[[outcome_var]])) +
      geom_boxplot(
        outlier.alpha = 0.25,
        outlier.colour = COL_NEUTRAL,
        fill = COL_PRIMARY,
        colour = COL_NEUTRAL,
        linewidth = 0.4
      ) +
      labs(
        x = label_from_map(v),
        y = label_from_map(outcome_var)
      )
  }
}

```

```

    ) +
    theme_project()

    save_plot(paste0("04_response_by_", v, ".png"), p_cat, width = 7.5, height = 5.5)
  }
}

# ---- EDA summary -----

mean_y <- mean(df[[outcome_var]], na.rm = TRUE)
var_y <- var(df[[outcome_var]], na.rm = TRUE)

# record dispersion diagnostics for model justification
write_lines_safe(
  c(
    paste0("EDA run: ", format(Sys.time())),
    "",
    paste0("N songs: ", nrow(df)),
    "",
    paste0("Outcome mean: ", round(mean_y, 2)),
    paste0("Outcome variance: ", round(var_y, 2)),
    paste0("Dispersion ratio (Var/Mean): ", round(var_y / mean_y, 2)),
    "",
    paste0("Share of songs lasting 1 week: ", round(100 * mean(df[[outcome_var]] == 1), 1),
    "%"),
    paste0("Share of songs lasting <= 2 weeks: ", round(100 * mean(df[[outcome_var]] <= 2),
    1), "%"),
    ),
  "outputs/descriptives/eda_summary.txt"
)

invisible(TRUE)
}

# ---- Run -----

song_df <- read_csv("clean/song_df.csv", show_col_types = FALSE)
song_df <- encode_factors(song_df)

run_eda(
  df = song_df,
  outcome_var = OUTCOME_VAR,
  core_vars = core_vars,
  audio_vars = audio_vars,
  cat_vars = cat_vars,
  n_bins = 25

```



```
)
```

```
# End of file -----
```

03_fit_nb_models.R

```
# R/03_fit_nb_models.R -----
# Fit count models for chart longevity (weeks_on_chart).
# Input: clean/song_df.csv
# Output: outputs/models/all_models.rds; outputs/models/model_fit_meta.rds

suppressPackageStartupMessages({
  library(dplyr)
  library(tidyr)
  library(readr)
  library(MASS) # glm.nb (masks dplyr::select())
  library(AER) # dispersiontest
})

source("R/analysis_setup.R")
source("R/helpers.R")

# ---- Output paths -----

DIR_MODELS <- "outputs/models"
PATH_MODELS <- file.path(DIR_MODELS, "all_models.rds")
PATH_META <- file.path(DIR_MODELS, "model_fit_meta.rds")

ensure_dirs(DIR_MODELS)

# ---- Model formulas -----

# Main model used for reporting (core + acoustic features)
f_main <- stats::as.formula(
  paste0(OUTCOME_VAR, " ~ ", paste(c(core_vars, audio_vars), collapse = " + "))
)

# Comparison model including musical categoricals
f_cat <- stats::update(
  f_main,
  stats::as.formula(paste(". ~ . +", paste(cat_vars, collapse = " + ")))
)
```

```
# Poisson is fitted for overdispersion comparison on the richest specification
f_pois <- f_cat
```

```
# ---- Data -----
```

```
song_df <- read_csv("clean/song_df.csv", show_col_types = FALSE)
song_df <- encode_factors(song_df)
```

```
model_vars <- all.vars(f_pois)
```

```
model_df <- song_df |>
  dplyr::select(all_of(model_vars)) |>
  drop_na()
```

```
std <- standardise_predictors(
  df = model_df,
  core_vars = core_vars,
  audio_vars = audio_vars,
  exclude = "n_artists_collab"
)
```

```
model_df_scaled <- std$df_scaled
scale_vars <- std$scale_vars
```

```
# ---- Design check (richest specification) -----
```

```
X <- stats::model.matrix(f_pois, data = model_df_scaled)
design_rank <- qr(X)$rank
design_p <- ncol(X)
```

```
# ---- Fit models -----
```

```
m_pois <- stats::glm(
  f_pois,
  data = model_df_scaled,
  family = stats::poisson(link = "log")
)
```

```
disp_test <- AER::dispersiontest(m_pois)
```

```
m_nb_main <- MASS::glm.nb(f_main, data = model_df_scaled)
m_nb_cat <- MASS::glm.nb(f_cat, data = model_df_scaled)
```

```
models <- list(
  poisson = m_pois,
```

```

nb_main = m_nb_main,
nb_with_categoricals = m_nb_cat
)

# ---- Save models and metadata -----

save_rds_safe(models, PATH_MODELS)

fit_meta <- list(
  run_time = Sys.time(),
  n = nrow(model_df_scaled),
  dropped_na_n = nrow(song_df) - nrow(model_df_scaled),
  scale_vars = scale_vars,
  design_rank = design_rank,
  design_p = design_p,
  poisson_overdispersion = list(
    statistic = unname(disptest$statistic),
    p_value = unname(disptest$p.value)
  )
)

save_rds_safe(fit_meta, PATH_META)

message("Model fitting completed.")

```

04_report_nb_outputs.R

```

# R/04_report_nb_outputs.R -----
# Reporting outputs for Negative Binomial modelling of song longevity.
#
# Inputs:
# - outputs/models/all_models.rds
# - outputs/models/model_fit_meta.rds
# - clean/song_df.csv
#
# Outputs:
# - outputs/models/model_summary.txt
# - outputs/models/final_model_predictor_summary_continuous.csv
# - outputs/figures/nb_dharma_residuals.png
# - outputs/figures/nb_effects_percent_significant_only.png
#
# Notes:

```

```
# - nb_main is used for reporting; nb_with_categoricals is used for comparison.  
# - Continuous predictors are standardised in fitting (03); effects are rescaled  
# to raw-unit increments using the complete-case modelling sample.  
#
```

```
suppressPackageStartupMessages({  
  library(dplyr)  
  library(readr)  
  library(stringr)  
  library(forcats)  
  library(ggplot2)  
  library(performance) # check_collinearity  
  library(broom)      # tidy()  
  library(DHARMA)     # simulateResiduals  
  library(tidyr)  
  library(tibble)  
})
```

```
source("R/analysis_setup.R")  
source("R/helpers.R")
```

```
# ---- Parameters + dirs -----
```

```
DHARMA_N    <- 1000  
DHARMA_SEED <- 123  
IRR_P_CUTOFF <- 0.05
```

```
DIR_MODELS <- "outputs/models"  
DIR_FIGURES <- "outputs/figures"  
ensure_dirs(c(DIR_MODELS, DIR_FIGURES))
```

```
# ---- Load models + metadata -----
```

```
models <- readRDS(file.path(DIR_MODELS, "all_models.rds"))  
fit_meta <- readRDS(file.path(DIR_MODELS, "model_fit_meta.rds"))
```

```
required_models <- c("poisson", "nb_main", "nb_with_categoricals")  
missing_models <- setdiff(required_models, names(models))  
if (length(missing_models) > 0) {  
  stop(  
    "Missing expected models in all_models.rds: ",  
    paste(missing_models, collapse = ", "),  
    call. = FALSE  
  )  
}
```

```

m_pois <- models$poisson
m_nb <- models$nb_main
m_nb_c <- models$nb_with_categoricals

m_nb_null <- update(m_nb, . ~ 1)

# ---- Helpers -----

tidy_irr <- function(m, conf_level = 0.95) {
  broom::tidy(m, conf.int = TRUE, conf.level = conf_level) |>
    dplyr::filter(term != "(Intercept)") |>
    dplyr::mutate(
      IRR = exp(estimate),
      IRR_low = exp(conf.low),
      IRR_high = exp(conf.high)
    ) |>
    dplyr::select(term, estimate, std.error, statistic, p.value, IRR, IRR_low, IRR_high)
}

p_txt <- function(p) {
  if (p < 0.001) "p < 0.001" else paste0("p = ", signif(p, 3))
}

extract_lrt <- function(a) {
  cn <- trimws(colnames(a))
  colnames(a) <- cn
  list(
    df = as.integer(a[2, "df"]),
    stat = as.numeric(a[2, "LR stat."]),
    p = as.numeric(a[2, "Pr(Chi)"])
  )
}

# ---- Complete-case raw sample (SD conversion + descriptives) -----

song_df_raw <- read_csv("clean/song_df.csv", show_col_types = FALSE) |>
  encode_factors()

model_vars_rich <- all.vars(stats::formula(m_pois))

df_cc_raw <- song_df_raw |>
  dplyr::select(dplyr::all_of(model_vars_rich)) |>
  drop_na()

num_vars_raw <- names(df_cc_raw)[vapply(df_cc_raw, is.numeric, logical(1))]
num_vars_raw <- setdiff(num_vars_raw, OUTCOME_VAR)

```

```

scale_vars_fit <- unique(as.character(fit_meta$scale_vars))

sd_lookup <- df_cc_raw |>
  summarise(across(all_of(scale_vars_fit), sd)) |>
  pivot_longer(everything(), names_to = "term", values_to = "sd_raw")

# ---- IRR table objects (NB main) -----

nb_irr_full <- tidy_irr(m_nb)

nb_irr_txt <- nb_irr_full |>
  mutate(
    b = round(estimate, 3),
    SE = round(std.error, 3),
    IRR = round(IRR, 3),
    IRR_low = round(IRR_low, 3),
    IRR_high = round(IRR_high, 3),
    p_txt = format.pval(p.value, digits = 3, eps = 0.001)
  ) |>
  dplyr::select(term, b, SE, IRR, IRR_low, IRR_high, p_txt)

irr_hdr <- sprintf(
  "%-36s %7s %7s %8s %8s %8s %10s",
  "term", "b", "SE", "IRR", "low", "high", "p"
)

irr_rows <- sprintf(
  "%-36s %7.3f %7.3f %8.3f %8.3f %8.3f %10s",
  nb_irr_txt$term,
  nb_irr_txt$b,
  nb_irr_txt$SE,
  nb_irr_txt$IRR,
  nb_irr_txt$IRR_low,
  nb_irr_txt$IRR_high,
  nb_irr_txt$p_txt
)

irr_lines <- c(
  "----- IRR table (NB main; 95% CI) -----",
  "",
  irr_hdr,
  strrep("-", nchar(irr_hdr)),
  irr_rows,
  ""
)

```

```

# ---- Model summary (text-only) -----

lrt_cat <- suppressWarnings(anova(m_nb, m_nb_c, test = "LRT"))
lrt_cat <- extract_lrt(lrt_cat)

lrt_line <- paste0(
  "LRT (NB main vs NB + categoricals): \u03C7\u00B2(",
  lrt_cat$df, ") = ",
  signif(lrt_cat$stat, 4),
  ", ",
  p_txt(lrt_cat$p)
)

lrt_omni <- suppressWarnings(anova(m_nb_null, m_nb, test = "LRT"))
lrt_omni <- extract_lrt(lrt_omni)

omni_line <- paste0(
  "Omnibus LRT (NB main vs null): \u03C7\u00B2(",
  lrt_omni$df, ") = ",
  signif(lrt_omni$stat, 4),
  ", ",
  p_txt(lrt_omni$p)
)

design_rank <- fit_meta$design_rank
design_p <- fit_meta$design_p
n_obs <- fit_meta$n

scaled <- unique(as.character(fit_meta$scale_vars))
cont_all <- intersect(c(core_vars, audio_vars), names(stats::model.frame(m_nb)))
not_scaled <- setdiff(cont_all, scaled)

od <- fit_meta$poisson_overdispersion
overdisp_line <- paste0(
  "Dispersion test (Poisson): statistic = ",
  signif(od$statistic, 4),
  ", ",
  p_txt(od$p_value)
)

ic_tbl <- tibble::tibble(
  Model = c("Poisson", "NB main (frozen)", "NB + categoricals"),
  AIC = c(AIC(m_pois), AIC(m_nb), AIC(m_nb_c)),
  BIC = c(BIC(m_pois), BIC(m_nb), BIC(m_nb_c))
)

```

```

ic_lines <- c(
  "----- Model comparison (AIC/BIC; lower is better) -----",
  "",
  sprintf("%-28s %12s %12s", "Model", "AIC", "BIC"),
  strrep("-", 56),
  sprintf("%-28s %12.2f %12.2f", ic_tbl$Model, ic_tbl$AIC, ic_tbl$BIC)
)

cook <- stats::cooks.distance(m_nb)
cook_thr <- 4 / length(cook)

cook_max <- max(cook, na.rm = TRUE)
cook_n_above <- sum(cook > cook_thr, na.rm = TRUE)

ord <- order(cook, decreasing = TRUE, na.last = NA)
top_idx <- head(ord, 5)
top_pairs <- if (length(top_idx) == 0) "<none>" else {
  paste0(top_idx, ":", signif(cook[top_idx], 4), collapse = ", ")
}

lines <- c(
  "=====",
  paste0("Report run: ", format(Sys.time())),
  "=====",
  "",
  "",
  "----- Sample size / design matrix (richest spec used in fitting) -----",
  paste0("N = ", n_obs),
  paste0("rank(X) = ", design_rank, " ; ncol(X) = ", design_p),
  if (design_rank < design_p) {
    "WARNING: rank deficiency detected (perfect collinearity / aliasing likely)."
  } else {
    "OK: full rank design matrix."
  },
  "",
  "----- Scaling -----",
  paste0("Scaled (1 SD): ", paste(scaled, collapse = ", ")),
  paste0("Not scaled: ", if (length(not_scaled) == 0) "<none>" else paste(not_scaled, collapse =
  ", ")),
  "",
  "----- Omnibus model test -----",
  omni_line,
  "",
  "----- Poisson overdispersion test -----",
  overdisp_line,

```



```

    """,
    ic_lines,
    """,
    "----- Likelihood ratio test (nested NB models) -----",
    lrt_line,
    """,
    "----- Collinearity (NB main) -----",
    paste(capture.output(performance::check_collinearity(m_nb)), collapse = "\n"),
    """,
    "----- Influence (Cook's distance; NB main) -----",
    paste0("Rule-of-thumb threshold: 4/n = ", signif(cook_thr, 4)),
    paste0("Max Cook's distance: ", signif(cook_max, 4)),
    paste0(
      "N above threshold: ",
      cook_n_above,
      " (",
      round(100 * cook_n_above / n_obs, 1),
      "%)"
    ),
    paste0("Top Cook's distances (index:value): ", top_pairs),
    """,
  )

lines <- c(
  lines,
  "----- Model identity check -----",
  paste0("fit_meta run_time: ", fit_meta$run_time),
  paste0("nobs(m_nb): ", stats::nobs(m_nb)),
  paste0("formula(m_nb): ", paste(deparse(stats::formula(m_nb)), collapse = " ")),
  """,
)

lines <- c(lines, irr_lines)

write_lines_safe(lines, file.path(DIR_MODELS, "model_summary.txt"))

# ---- Continuous predictor summary (raw distribution + scaling) -----

predictor_summary_continuous <- df_cc_raw |>
  dplyr::select(dplyr::all_of(num_vars_raw)) |>
  summarise(
    across(
      everything(),
      list(
        mean = mean,
        sd   = sd,

```

```

    min  = min,
    max  = max,
    median = median,
    q25  = ~ quantile(.x, 0.25, na.rm = TRUE),
    q75  = ~ quantile(.x, 0.75, na.rm = TRUE),
    p10  = ~ quantile(.x, 0.10, na.rm = TRUE),
    p90  = ~ quantile(.x, 0.90, na.rm = TRUE)
  ),
  .names = "{.col}__{.fn}"
)
)|>
pivot_longer(
  cols = everything(),
  names_to = c("term", "stat"),
  names_sep = "__",
  values_to = "value"
)|>
pivot_wider(names_from = stat, values_from = value) |>
mutate(IQR = q75 - q25) |>
left_join(INC_MAP, by = "term") |>
mutate(
  inc_over_SD = inc / sd,
  inc_over_IQR = inc / IQR
)|>
arrange(term)

write_csv(
  predictor_summary_continuous,
  file.path(DIR_MODELS, "final_model_predictor_summary_continuous.csv")
)

# ---- Figures -----

# DHARMA is used to check calibration / dispersion / outliers post-fit.
set.seed(DHARMA_SEED)
sim <- DHARMA::simulateResiduals(m_nb, n = DHARMA_N)

out_resid <- file.path(DIR_FIGURES, "nb_dharma_residuals.png")
grDevices::png(out_resid, width = 1400, height = 1000, res = 200)
plot(sim)
grDevices::dev.off()

# Significant-term effects are rescaled to raw-unit increments for interpretability.
sig <- nb_irr_full |>
  filter(!is.na(p.value), p.value < IRR_P_CUTOFF) |>
  mutate(

```

```

group = case_when(
  term %in% c(
    "song_rank_entry",
    "artist_prior_n_songs",
    "artist_avg_longevity_prior",
    "artist_avg_peak_rank_prior"
  ) ~ "Pre-entry factors",
  term %in% audio_vars ~ "Acoustic features",
  TRUE ~ "Controls"
),
group = factor(group, levels = c("Pre-entry factors", "Acoustic features", "Controls")),
label = paste0(label_from_map(term), label_increment_suffix(term))
) |>
left_join(INC_MAP, by = "term") |>
mutate(inc = if_else(is.na(inc), 1, inc)) |>
left_join(sd_lookup, by = "term") |>
mutate(
  expo = case_when(
    term %in% scale_vars_fit ~ inc / sd_raw,
    TRUE ~ inc
  ),

  log_irr    = log(IRR),
  log_irr_low = log(IRR_low),
  log_irr_high = log(IRR_high),

  irr_inc    = exp(log_irr    * expo),
  irr_low_inc = exp(log_irr_low * expo),
  irr_high_inc = exp(log_irr_high * expo),

  pct    = (irr_inc - 1) * 100,
  pct_low = (irr_low_inc - 1) * 100,
  pct_high = (irr_high_inc - 1) * 100,

  ord = abs(pct)
) |>
group_by(group) |>
mutate(label = forcats::fct_reorder(label, ord)) |>
ungroup() |>
arrange(group, desc(ord)) |>
mutate(
  x_lab = if_else(pct >= 0, pct_high + 1.2, pct_low - 1.2),
  hjust = if_else(pct >= 0, 0, 1),
  pct_label = sprintf("%+.0f%%", pct)
)

```

```

x_lim <- max(abs(c(sig$pct_low, sig$pct_high)), na.rm = TRUE) + 3

p_sig <- ggplot(sig, aes(y = label)) +
  geom_vline(xintercept = 0, linetype = "dashed", linewidth = 0.8, colour = COL_REF_LINE)
+
  geom_errorbar(
    aes(xmin = pct_low, xmax = pct_high),
    orientation = "y",
    height = 0.18,
    linewidth = 0.7,
    colour = "grey55"
  ) +
  geom_point(aes(x = pct), size = 3.0, colour = COL_PRIMARY) +
  geom_text(
    aes(x = x_lab, label = pct_label, hjust = hjust),
    size = 3.6,
    colour = COL_NEUTRAL
  ) +
  facet_grid(group ~ ., scales = "free_y", space = "free_y", switch = "y") +
  scale_x_continuous(
    limits = c(-x_lim, x_lim),
    expand = expansion(mult = c(0.02, 0.08))
  ) +
  labs(
    x = "Percentage change in expected chart longevity",
    y = NULL
  ) +
  theme_project() +
  theme(
    panel.grid.minor = element_blank(),
    panel.grid.major.y = element_blank(),
    strip.placement = "outside",
    strip.background = element_rect(fill = "grey92", colour = NA),
    strip.text.y.left = element_text(angle = 90, face = "bold", size = 12),
    axis.text.y = element_text(size = 10),
    plot.margin = margin(10, 16, 12, 10)
  )

ggsave(
  filename = file.path(DIR_FIGURES, "nb_effects_percent_significant_only.png"),
  plot = p_sig,
  width = 9.5,
  height = 7.2,
  units = "in",
  dpi = 300,
  bg = "white"
)

```

```
)
```

```
message("Reporting complete. Outputs written to outputs/models and outputs/figures.")
```

Appendix B

GitHub Project and Project Screenshots

Figure B1

GitHub Profile

The screenshot shows Archie Irving's GitHub profile. On the left, there's a circular profile picture of a young man with curly hair. Below it, the name 'Archie' and username 'ArchieIrving' are displayed, along with an 'Edit profile' button. It shows 8 followers and 4 following, and a location of 'England'. Under 'Achievements', there are two badges: a GitHub Star badge and a 'New!' badge. Under 'Organizations', there is a small square icon. The main content area on the right has a header 'Hi, I'm Archie Irving' with a hand icon. Below this is a bio: 'MSc Data Science Student | BSc Psychology (First-Class, 89%)'. A paragraph follows: 'I specialise in behavioural data, statistical modelling, and applied analytics. My background in psychology and data science allows me to understand human behaviour through a quantitative lens and translate insights into real-world applications.' Another paragraph states: 'Outside of formal education, I have applied psychology and data science tools to build and manage small businesses, most recently a clothing brand (badmannaz.com). This work involved analysing customer behaviour and marketing performance to support data-driven decisions across strategy, branding, and operations.' The 'Projects' section lists two items: 'MSc Data Science Project' with a link to 'Redefining Musical Success: Chart Longevity' and a description, and 'Undergraduate Dissertation (BSc Psychology)' with a link to 'Evaluating the Clinical Validity of ADHD Portrayals on TikTok' and a description. A 'Fun Fact' section mentions programming as a teenager while running a Minecraft server. The 'Connect With Me' section lists LinkedIn and Business Website links. At the bottom, a 'Archie's GitHub Stats' box shows 1 Total Stars Earned, 243 Total Commits, and 42 Total PRs, accompanied by a GitHub logo.

ArchieIrving / README.md

Hi, I'm Archie Irving

MSc Data Science Student | BSc Psychology (First-Class, 89%)

I specialise in behavioural data, statistical modelling, and applied analytics. My background in psychology and data science allows me to understand human behaviour through a quantitative lens and translate insights into real-world applications.

Outside of formal education, I have applied psychology and data science tools to build and manage small businesses, most recently a clothing brand (badmannaz.com). This work involved analysing customer behaviour and marketing performance to support data-driven decisions across strategy, branding, and operations.

Projects

- MSc Data Science Project**
[Redefining Musical Success: Chart Longevity](#)
An analysis of musical success using chart longevity as an outcome, examining the role of artist history, exposure, and song-level characteristics through statistical modelling.
- Undergraduate Dissertation (BSc Psychology)**
[Evaluating the Clinical Validity of ADHD Portrayals on TikTok](#)
A quantitative content analysis examining the alignment between social media portrayals of ADHD and established clinical diagnostic criteria.

Fun Fact

I began programming as a teenager while running a Minecraft server, which led to a long-term role as a developer and head game designer on ProjectKorra. Over 10+ years, this work involved software development, database-backed systems (SQL/MySQL), team leadership, system and economy design, and data-driven decision-making across analytics, SEO, marketing, and e-commerce. This experience strongly shaped my interest in user behaviour, system design, and modelling, ultimately leading me toward data science.

Connect With Me

- LinkedIn:** <https://www.linkedin.com/in/archie-irving-02a445103>
- Business Website:** <https://badmannaz.com>

Archie's GitHub Stats

★ Total Stars Earned:	1
🕒 Total Commits:	243
🔗 Total PRs:	42

Figure B2

GitHub Project Landing Page

 hit-song-science Public

Pin Watch 0

main 1 Branch 0 Tags Add file Code

 ArchieIrvig

Refactor data cleaning script for clarity and efficiency

7576157 · yesterday 40 Commits

R	Refactor data cleaning script for clarity and efficiency	yesterday
clean	Delete clean/.gitkeep	2 days ago
outputs	Delete outputs/models/.gitkeep	2 days ago
raw	Delete raw/popularity/ReadMe.txt	2 days ago
README.md	Update README.md	2 days ago

Figure B3

Project [README.md](#) Available on the GitHub Project Page

Modelling Song Chart Longevity

Project Overview

The aim of this project was to extend research in *Hit Song Science* by adopting an explanatory modelling approach to musical success. Rather than treating success as a binary outcome (chart entry versus non-entry), the study operationalised success as chart longevity, measured by the number of weeks a song remained on the *Billboard* charts.

Data and Methods

The analysis used data from MusicOSet, an open, curated dataset integrating *Billboard* chart performance with song-level metadata and Spotify-derived acoustic features. The dataset contains 20,405 songs from the U.S. popular music industry spanning 1962–2018.

To examine sustained chart presence, a Negative Binomial regression model was estimated to address two research questions:

- **RQ1:** To what extent do artist history, prior popularity, and exposure-related factors explain sustained chart presence?
- **RQ2:** What role do song-level characteristics play in shaping sustained chart presence, conditional on artist and exposure factors?

Key Findings

The results indicate that artist prior chart longevity is the strongest positive predictor of weeks on chart, while entry rank exhibits a smaller but positive association. In contrast, prior peak chart position and repeated chart exposure are negatively associated with chart longevity, suggesting diminishing returns to peak-oriented or repeated success.

Song-level acoustic features were statistically significant but exhibited **modest effect sizes**. Higher danceability and loudness were positively associated with chart longevity, whereas greater speechiness, acousticness, liveness, instrumentalness, and energy were negatively associated. Overall, the findings reinforce long-standing conclusions in *Hit Song Science* that audio features play a limited explanatory role relative to artist history and exposure dynamics.

How to Run the Analysis

Requirements

- R (version 4.x)
- R packages: tidyverse, MASS, AER, performance, DHARMa, broom

How to Run the Analysis

1. Clone or download this repository to your local machine and open the project in R or RStudio.
2. Set the project root directory as your working directory. The project root is the top-level folder containing the `R/`, `raw/`, `clean/`, and `outputs/` directories.
3. Ensure all required R packages are installed (see the package loading statements at the top of the scripts).
4. Run the full analysis pipeline from the project root using:

```
source("R/00_run_all.R")
```



Running `00_run_all.R` executes the entire analysis pipeline end-to-end. This includes data cleaning and aggregation.

